2 PBW at CIBA

In this chapter, we review the layout of the proton beam writing at the Centre of Ion Beam Applications (CIBA), NUS. Figure 2-1 shows the layout of the beamlines in CIBA. A 3.5 MV Singletron accelerator from HVEE is used for 5 different beam lines shown in the schematic.

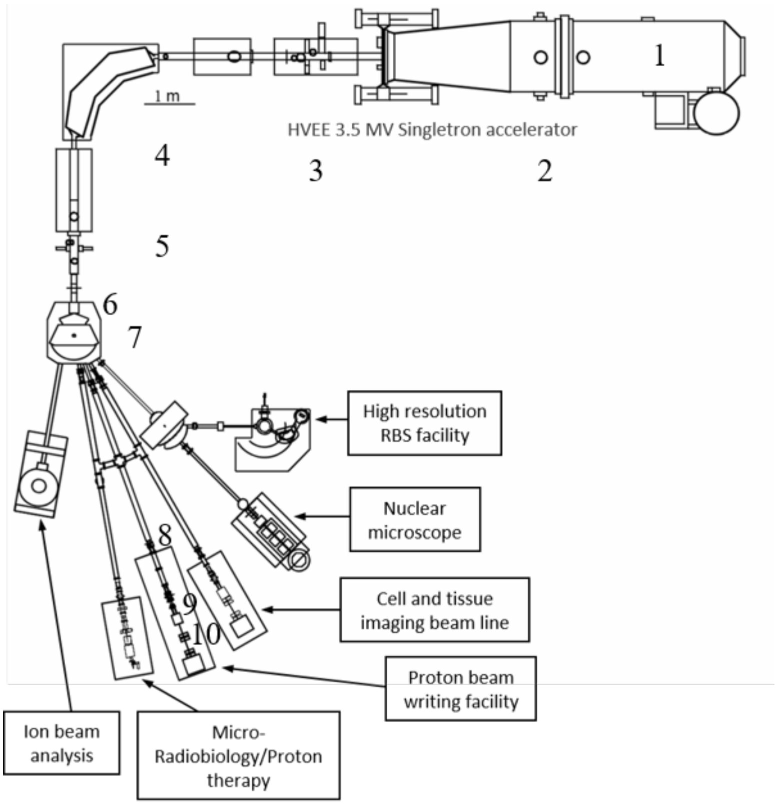

Figure 2-1. Accelerator and beamline layout at Centre for Ion Beam Applications (CIBA), NUS

2.1 Accelerator layout

Ions generated by exciting incoming gas to a source is extracted by applying electric field across the source (2). Energy of the ions is defined by the terminal voltage applied (1). Ion sources generate a variety of ion species of different charge states. For example, hydrogen gas can be easily excited to generate molecular hydrogen ions with a +1-charged state or protons with +1 charge state. The required charge (Q) and masses (M) are filtered by the 90° magnet (4) which filters out the non-relevant ion species with the resolution of Q/M. By placing this magnet appropriately, it can also be used to focus the beam at the objective slit plane. Steering (3) of the ion beam out of the accelerator is controlled by applying electric fields in orthogonal directions to optimize passage of a maximized current and aligned beam. The objective aperture (5) is used to define the virtual source size of the beam and to reduce the beam divergence from the accelerator. A switcher magnet (7) bends the beam towards the end station by applying the proportional magnetic field. Blanking plates (6) are placed before the switching magnet to deflect the beam away from the beam line which stops the flow of ions towards the target station. A de-magnified image of the objective aperture is formed on the target plane by the quadrupole lens focusing system (10). For nanometre spot sizes, controlling the divergence before the focusing lenses is important. This is done by using collimator aperture (8) which reduces the beam divergence for the finer focusing through the lenses at the expense of losing beam current. The beam is scanned on the target plane by scanners (9) placed before or after quadrupole lenses.

2.2 Beam defining apertures

Beam defining slits are used as objective and collimating apertures to shape the beam in nuclear microprobes. MeV ions are stopped within these apertures due to their higher density and thickness, greater than the end of the range of ions in materials (<10 μm). The non-interacting part of the ion beam passes through the aperture openings which can then be focused to a spot on the target plane using electric or magnetic lenses. Spot size of the focused beam in such systems is defined by demagnified aperture size in the objective plane [125]. The collimator apertures are used to reduce the divergence and aberrations of the beam drifting from the objectives upon scanning through the quadrupole lenses [126][127]. The effect of scattering from the slits and impact on beam focusing was studied by Gorelick *et. al.* [128] and experimentally reported by van Kan *et. al.* [129] and Watt *et. al.* [130]. Some ions passing through the objective slits are scattered at the edges (i.e. transparency zone) leading to an increased angular spread of the beam along with an increased energy spread.

In the nuclear microprobe systems at Centre for Ion Beam Applications (CIBA), tungsten carbide (WC) slits are used for defining beam centroid in objective and collimator apertures. These slits serve for applications ranging from material characterization using high resolution Rutherford Back Scattering (RBS) [131], proton beam writing (PBW) [59], [129] and bio-imaging [132], [133] using focused ion beams. The slits are attached to the end of a manually operated differential micrometre to define the aperture size and alignment (Mitutoyo 110-101).

2.3 Electrostatic scanners

Electrostatic scanner plates in orthogonal direction are placed before the quadrupole lenses to scan the beam in image plane. Scanners are driven by individual amplifiers depending on the appropriate scan size. Larger scan sizes are achieved using Trek amplifier (Trek 609E-6 ±4 kV) with a noise level of 50mV, and smaller scan sizes are achieved using Trek amplifiers with noise levels of 70 μV. The ratio of the scan output voltage follows the ratio of demagnification indicating a stigmatic system.

2.4 Beam blanking

A Fischer amplifier (dual ±220V High Voltage) with a slew rate of 1000 V/μs is placed before the switching magnet that applies high voltage across two parallel plates and deflects the beam away in the horizontal direction. This allows for fabrication of arbitrary structures during beam scanning.

2.5 Probe-forming lenses

Due to higher magnetic rigidity of ions accelerated to MeV energies, quadrupole lenses with radial distribution of fields are needed to focus to smaller spot size.

Lorentz force acting on a particle with charge q, and velocity v in a magnetic field B can be written as:

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \tag{2.1}$$

Magnetic field in quadrupoles are generated by the current running through the coils wrapped around the magnetic pole pieces. Figure 2-2 shows the cross-sectional view of a quadrupole magnet.

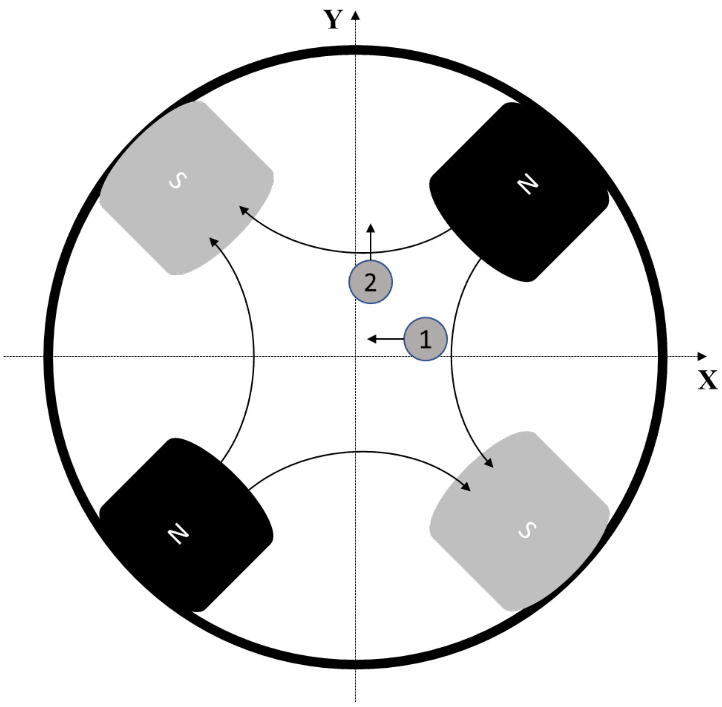

Figure 2-2. Cross-section of a quadrupole lens. Shown are the four pole pieces generating magnetic field lines in the space within. Positively charged particles at position 1 and 2 travel into the plane of the paper

Magnetic field lines are contained within the lenses and are orthogonal to the incoming charged particle travelling into the plane of the paper. A positively charged particle at position 1 will experience a force towards the centre, whereas particles at position 2 will experience an outward force. A single quadrupole lens will focus the beam in one plane and defocus in orthogonal plane. Therefore at least two quadrupole lenses are required for defining a spot focus.

A few assumptions can be considered for quadrupole lenses: They are long enough to avoid components of magnetic field in the direction of beam. Field is calculated in current free region. By Ampere's law,

$$\vec{\nabla} \times \vec{\mathbf{B}} = 0 \qquad (2.2)$$

This implies a scalar potential, V such that $\vec{\boldsymbol{B}} = \boldsymbol{\nabla} V$. Also, divergence of magnetic field is zero, i.e. $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$. This implies

$$\nabla^2 V = 0 \tag{2.3}$$

This Laplace equation can be solved by four-fold symmetric scalar potential of the form

$$\phi(r, \theta) = \sum_{n=2,6,10\ldots} k_{2n}(\sin n\theta) r^n \tag{2.4}$$

However, due to practical limitations in manufacturing of the pole pieces, higher order harmonic terms need to be considered. The general expression of the scalar potential could then be written as

$$\phi(r, \theta) = \sum_{n=2}^{\infty} \sum_{m=2}^{\infty} v_{mn} k_{mn} r^n \sin(n\theta - \alpha_{mn}) \tag{2.5}$$

Where n is the multipole index and m is the boundary condition (equation 2.6) in the solution of Laplace equation

$$\phi(r, \theta) = -\phi\left(r, \theta + \frac{\pi}{m}\right) \tag{2.6}$$

$$v_{mn} = \begin{cases} 1, & n = m, 3m, 5m \\ 0, & \text{otherwise} \end{cases} \tag{2.7}$$

Solving for **B** in cartesian coordinates with magnetic plane described in XY plane, the strength of the magnetic field in X direction is dependent on the distance from Y axis, while field in Y direction is dependent on the relative position from X axis as

$$B_x = GY \tag{2.8}$$

$$B_y = GX \tag{2.9}$$

where, G is the magnetic field gradient and given by

$$G \backsimeq \frac{2\mu_0 nI}{r_0^2} \tag{2.10}$$

Multiple lens system:

A typical nuclear microprobe beam line is an inverted microscope where the object aperture is demagnified by a set of magnetic quadrupole lenses to form an image at the target. Various configurations can be defined based on the number of lenses used namely doublet, triplet, quadruplets and so on. Higher demagnification has been shown using triplets and spaced Oxford triplet configurations at the expense of asymmetric demagnifications and chromatic and spherical aberrations. Quadruplets offer symmetric demagnification with lower spherical aberration at the expense of lower demagnification and need of higher stability of power supplies. A typical triplet configuration is called a converging – diverging and converging configuration with a cross over in X direction. Spacing the first two lenses in the direction of beam can further increase the demagnification in the converging direction.

Beam coordinates in the objective plane can be represented by a vector $\vec{x}_o$ which is a N×1 matrix. Such that $\vec{x}_o \equiv [x_o, y_o, \theta_o, \phi_o, \delta]$.

Here, x_o, y_o are the position coordinate of an ion in the object XY plane, θ_o and ϕ_o are the projection angle of the travelling ion with respect XZ-plane and YZ-planes respectively. δ is the momentum dispersion of the ion.

For a single quadrupole lens, matrix method defines the focusing and defocusing effect in terms of a transform matrix M. Image coordinates, $\vec{x}_i$ as a result can be written in terms of $\vec{x}_o$ and M as

$$\vec{x}_i = M\vec{x}_o \tag{2.11}$$

For a set of multiple lenses, the image coordinate matrix can be defined in terms of P, the chain product of individual transfer matrices and a vector combination of initial coordinates expanded from each quadrupole element. Taking the first order transfer element for the system as illustrated by Grime *et. al.* [125], focal point in both X and Y directions should be independent of the divergences. This leads to a demagnification of the system in X and Y directions defined as

$$D_x = \langle x|x\rangle^{-1} \tag{2.12}$$

$$D_y = \langle y|y\rangle^{-1} \tag{2.13}$$

And image coordinates can be derived by expanding the matrix multiplication which defines the aberration of the system

$$x_i = A_o + A_1 x_o + A_2 y_o + A_3 \theta_o + \cdots + A_n v_1^i v_2^j v_3^k \tag{2.14}$$

Where v_m is the system variable and n (sum of the exponents i, j, k) describes the order of aberration.

2.6 Beam line optics

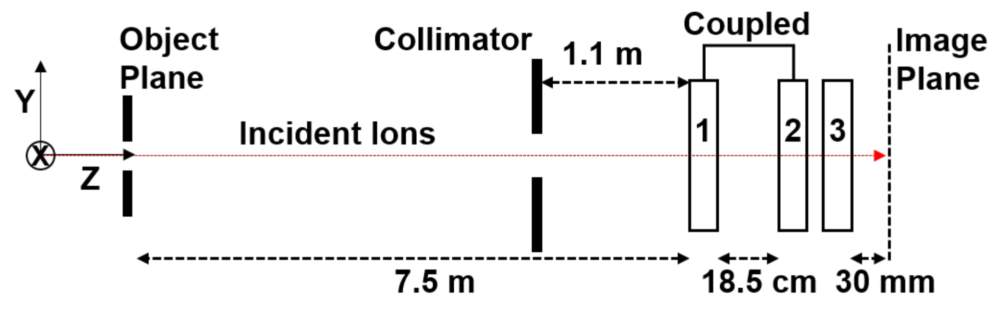

Figure 2-3. Layout of the beamline optics for PBW at CIBA. 3 Quadrupole lenses configured in spaced oxford triplet mode (CDC) with first and second lens coupled to the same power supply

Figure 2-3 shows the layout of the beam line optics used for Proton beam writing in CIBA. The optimization of the layout was done by van Kan *et. al.* [129] and Yao *et. al.* [134]. Proton beam writing system at CIBA uses spaced Oxford triplet configuration in converging-diverging-converging configuration. System parameters for this oxford triplet configuration are listed in Table 2-1 .

Table 2-1. System layout of spaced oxford triplet

Element	Distance
Object to Lens 1	7.5 m
Lens bore radius	3.75 mm
Lens length	55 mm
Spacing between lens 1 and 2	18.5 cm
Spacing between lens 2 and 3	2.5 cm
Image distance	30 mm

Beam optics simulations were carried out using particle beam optics laboratory 3.0 (PBO Lab) [135]. A beam of 2 MeV protons passes through a objective aperture set to 8 × 4 μm^2 with a half angle beam divergence of 3 μrad. Beam divergence is defined using the collimator aperture opening of 30 × 30 μm^2. Figure 2-4 bshows the spot focus attained with a demagnification of 857 × (-130). Table lists down the aberration coefficient for the system.

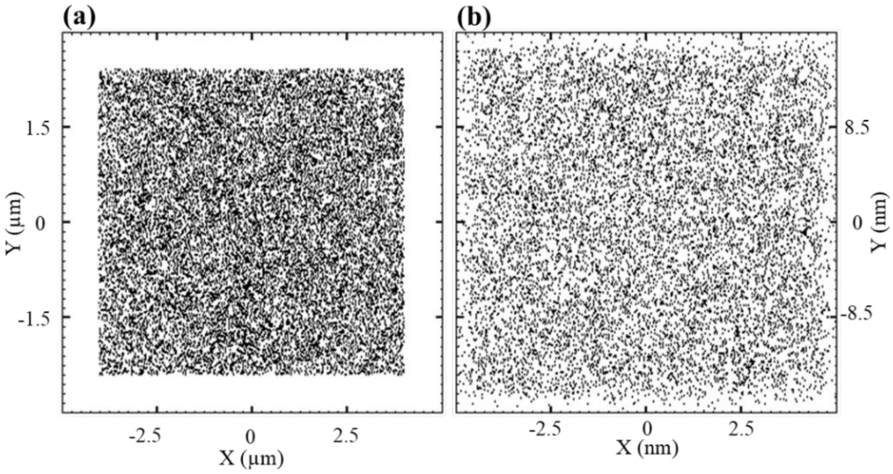

Figure 2-4. PBO Simulation of 2 MeV proton. The beam line demagnification is calculated to be 857 × 130. Figure (a) shows the particle distribution in an objective slit of opening 8 × 4 μm^2; (b) shows the beam spot size of 9.3 × 32 nm^2. Collimator aperture was set for 30 × 30 μm^2.

Table 2-2. Aberration Coefficient for spaced oxford triplet extracted from PBO simulations

Demagnification (First order)	$D_x = \langle x\|x\rangle^{-1}$	857
	$D_y = \langle y\|y\rangle^{-1}$	-130
Chromatic Aberration (Second order) (mm/mrad)	$\langle x\|\theta\delta\rangle^{-1}$	-52
	$\langle y\|\theta\delta\rangle^{-1}$	126
Spherical Aberration (Third order) mm/mrad^3	$\langle x\|\theta^3\rangle$	24
	$\langle x\|\theta\phi^2\rangle$	10.877
	$\langle y\|\phi^3\rangle$	-24
	$\langle y\|\theta^2\phi\rangle$	-71

Chromatic aberration of the ion beam reaching the probe forming lenses, triplets or quadruplets, limits the smallest spot size achievable. Typically, stability of 100 eV per MeV is required for resolution of less than a micron. Maintaining sub-10 nm stable spot sizes require a 10-ppm stability of the incident beam for probe forming lenses with higher demagnification. Singletron accelerator at CIBA has shown resolution down to 10 nm with 20 eV stability [134].

2.7 Beam size estimation and focusing performance

Estimation of the beam size is the key element to auto-focus and write features in photoresist. Beam is scanned across the edge of the grid and ions/electrons are collected. The collected line scan resembles a complimentary error function. This function can be modelled as a gaussian using central limit theorem. Figure 2-5 shows the schematic of a beam being scanned across the edge.

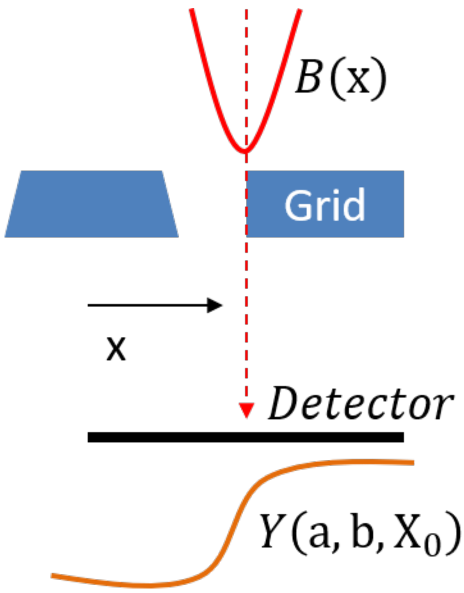

Figure 2-5 Schematic of beam spot estimation

The original beam can be modelled as 2-dimensional gaussian distribution independent of cross terms in XY as

$$B(x) = \frac{a}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-X_0)^2}{2\sigma^2}} \qquad (2.15)$$

Where a is the normalization factor to get the total beam current, X_0 is the beam centre which is scanned across the resolution standard edge and σ is the standard deviation of the beam current distribution. The grid bar can be defined as a step function, F(x) with edge positioned at b.

$$F(x) = \begin{cases} 1, & x < b \\ 0, & x \geq b \end{cases} \qquad (2.16)$$

Yield of the ions/electrons collected as a function of position in x direction when the beam is scanned across the grid edge, $Y(a,b,X_0)$ can be written as the integral of the product of the step function and the beam gaussian as

$$Y(a, b, X_0) = \int_{-\infty}^{\infty} F(x)B(x)dx \qquad (2.17)$$

Defining the step at x = b, and integrating equation

$$Y(a, b, X_0) = \int_{-\infty}^{b} B(x)dx \qquad (2.18)$$

$$Y(a, b, X_0) = \frac{a}{2}\left[1 + \operatorname{erf}\left(\frac{2\sqrt{\ln 2}}{f}(b - X_0)\right)\right] \qquad (2.19)$$

Where f is the full width half maximum and is related to the beam standard deviation σ as

$$f = 2\sigma\sqrt{2\ln 2} \approx 2.35\sigma \tag{2.20}$$

For the case of electron detection, an enhanced edge is detected owing to secondary electrons escaping both from the front surface and edge of the grid bar equation 2.19 is modified with an added beam distribution weighted as per the intensity described by

$$Y(a, b, X_0) = H_{low} + H_{err}\left(1 + \operatorname{erf}\left(\frac{2\sqrt{\ln 2}}{f}(b - X_0)\right)\right) + H_{gauss}e^{-\frac{\ln 16}{f^2}(b-X_0)^2} \tag{2.21}$$

From both equations 2.19 and 2.21, number of variables required to fit are 3 and 5 respectively. Table 2-3 below shows the percentage of the beam around the centre X_0 for different threshold.

Table 2-3 Total percentage of beam as a function of sigma

Beam region	Percentage
$b \pm \sigma$	16 to 84
$b \pm f/2$	12 to 88
$b \pm 2\sigma$	2.5 to 97.5

2.8 Target chamber

Samples are mounted inside a custom designed end station that houses a nano-positioning stage from PI using nexus piezo mechanism. Samples are positioned in the focal plane from the feedback of Omron interferometer (Omron ZS-HLDC11). Raster scans are generated from a PXI-6259 DAQ card from

National Instruments with an effective 13-bit resolution. A resolution standard is mounted on the sample holder to focus ion beams using forward scatter scanning transmission ion beam microscopy or direct STIM. Transmitted or forward scattered ions are collected using a PIN diode (Hamamatsu, S1223) reverse biased at 15 V bias. An annular multi-channel plate (NVT 2C45/C4M10) is mounted in front of the sample mounted on the chamber wall around the beam entrance to image ion induced secondary electrons. Ortec pre-amplifier (142A) and pulse shapers are used for STIM imaging and a custom designed pre-amplifier is used for imaging secondary electrons from MCP which converts current on the collector plate to voltage pulses with a gain of 20. The shaped pulses from both the signal processing system are fed to FAST ADC (dual ADC, 7072) and a single channel analyser. Noise level in the detection system is determined using the ADC spectrum. A TTL pulse is generated by the ADC/SCA corresponding to an ion/electron detection. This pulse is fed to the gate of the counter that records the counter status. A two-dimensional image is generated by converting the counter buffer to an intensity map as shown by Udalagama *et. al.* [136] Quadrupole currents are supplied from Bruker power supply with 2 ppm resolution. These power supplies are controlled through National Instruments VISA interface. Auto-focusing is done by defining the scan length across the edges of the resolution standard. Beam size is estimated using a complementary error function and non-linear fitting for Equations 2.19 and 2.21. Lithography is done using the same interface using step and repeat or stage scan combined with beam scan.

2.9 Data acquisition (automation)

An important part of the proton beam writing setup is focusing and estimating the beam spot size and using it for lithography on the samples mounted in the target chamber. Object and collimator slits are aligned to the quadrupole optical axis to minimize beam steerer while adjusting the lens current supply. Proton beam is then scanned over a free-standing resolution standard using electrostatic or magnetic scanners. As the beam scatters through the resolution standard, scattered particles are collected and imaged using particle detectors. Assuming a gaussian profile for the focused beam, a line scan fit is drawn while scanning the beam at the edges of the resolution standard in orthogonal directions. This routine can be automated by optimizing the quadrupole lens current supply while estimating the beam size. Imaging of resolution standard can be achieved using various techniques such as STIM (On and OFF axis), ion induced secondary electron emission, and Rutherford Backscatter Imaging. The minimum spot focus is defined based on the demagnification of the lenses limited by the aberrations. Once the spot focus is achieved, a writing file containing the geometry of the features to be written can be loaded and the data fed to the scanners. Blanking plates are placed after the object aperture to deflect the beam away from the target for the regions in the geometry where the beam is supposed to be off. Various strategies can be employed for writing structures. Scanning the beam over the defined area is limited by the beam clipping and quadrupole parasitic aberrations. Scan area is also limited by the beam broadening at the edges of the scan field. By combining a nano-positioning stage, a step and repeat procedure can be implemented which is however prone to stitching errors. Long waveguides and channels can be written with just moving the stage at target velocity corresponding to the dose required at a given beam current.

2.10 Summary

In this chapter, we introduced the implementation of PBW at CIBA. PBW beamline is based around a spaced-Oxford triplet magnetic quadrupole system used for focusing ions. This system achieves a demagnification of 857 × 130 and has demonstrated focusing capability down to 9.3 nm × 32 nm. We also discussed estimation of beam spot size using a two-dimensional gaussian model. The complimentary error function which represents the yield of ions or secondary electrons collected by the detector is then used to extract the beam FWHM. A brief description of the end station was provided along with the implementation of data acquisition and beam control.